

HDOC DAY -26

QUESTION:

Write a C program that displays a menu and uses a switch statement to print the “Right-aligned patterns” selected by the user. Use nested loops with arrays. Each pattern may be implemented in a separate parameter less function with no return value.

The patterns are:

1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle

Algorithm

1. Start.
2. Display the menu containing the three right-aligned pattern options.
3. Read the user’s choice.
4. Use a switch statement to select the required pattern.
5. Input the number of rows n.
6. Use nested loops to print the selected right-aligned pattern.
7. Display the pattern.
8. Stop.

Test Case Choice n			Expected Result
1	1	4	Right-aligned increasing * triangle
2	2	4	Right-aligned increasing number triangle
3	3	4	Right-aligned decreasing * triangle
4	4	—	Invalid choice

>C Program

```
#include <stdio.h>
```

```
void pattern1();
```

```
void pattern2();
```

```
void pattern3();
```

```
int main()
```

```
{
```

```
    int choice;
```

```
    printf("1. Right-aligned increasing triangle\n");
```

```
    printf("2. Right-aligned increasing number triangle\n");
```

```
    printf("3. Right-aligned decreasing triangle\n");
```

```
    printf("Enter your choice: ");
```

```
    scanf("%d", &choice);
```

```
    switch(choice)
```

```
    {
```

```
        case 1:
```

```
            pattern1();
```

```
            break;
```

```
        case 2:
```

```
            pattern2();
```

```
            break;
```

```
        case 3:
```

```
            pattern3();
```

```
            break;
```

```
        default:
```

```
            printf("Invalid choice\n");
```

```
    }
```

```
    return 0;
```

```
}
```

```
void pattern1()
```

```
{
```

```
    int n, i, j;
```

```
    printf("Enter number of rows: ");
```

```
    scanf("%d", &n);
```

```
    for(i = 1; i <= n; i++)
```

```
    {
```

```
        for(j = 1; j <= n - i; j++)
```

```
            printf(" ");
```

```
        for(j = 1; j <= i; j++)
```

```
            printf("* ");
```

```

        printf("\n");
    }
}

void pattern2()
{
    int n, i, j;

    printf("Enter number of rows: ");
    scanf("%d", &n);

    for(i = 1; i <= n; i++)
    {
        for(j = 1; j <= n - i; j++)
            printf(" ");

        for(j = 1; j <= i; j++)
            printf("%d ", j);

        printf("\n");
    }
}

void pattern3()
{
    int n, i, j;

    printf("Enter number of rows: ");
    scanf("%d", &n);

    for(i = n; i >= 1; i--)
    {
        for(j = 1; j <= n - i; j++)
            printf(" ");

        for(j = 1; j <= i; j++)
            printf("* ");

        printf("\n");
    }
}

```

>Compilation and execution of c program:

TEST CASE- 1,2,3,4

```
Session Actions Edit View Help
zsh: corrupt history file /home/priyadarshna/.zsh_history
priyadarshna@kali:~$ gcc day26.c -o day26
priyadarshna@kali:~$ ./day26
1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle
Enter your choice: 1
Enter number of rows: 4
      *
     **
    ***
   ****
priyadarshna@kali:~$ ./day26
1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle
Enter your choice: 2
Enter number of rows: 4
      1
     1 2
    1 2 3
   1 2 3 4
priyadarshna@kali:~$ ./day26
1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle
Enter your choice: 3
Enter number of rows: 4
   ***
  **
 *
priyadarshna@kali:~$ ./day26
1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle
Enter your choice: 4
Invalid choice
priyadarshna@kali:~$
```